H4 environmental-variable selection

Step 4: specify exploratory candidate equations and check full-design VIF/rank; no PCA/LASSO or automatic variable deletion.

	UCM			Atmo raw	Atmo average	LYS1 raw	LYS2 raw	LYS average	GVI imagery	SVF imagery	Sound			Solar	Empatica		Eye-tracker	Questionnaire	Metadata / QC			
STEP 1 **Inventory (226)**	AH AIR RH AIR T bot AIR T mid AIR T top AIR temp GPS speed kmh HUMIDEX HUMIDEX SR IO flag IR back IR down IR front IR left	IR right IR spot gnd IR spot left IR spot right IR spot sky IR up MRT MRT S OPT PET PET SR SND dBA SUN Gh SUN alt	SUN az UTCI UTCI SR WIND AWA WIND AWS WIND TWD WIND TWS fp	left humidity left pm1 left pm10 left pm2.5 left temperature right humidity right pm1 right pm10 right pm2.5 right temperature	pm1 avg pm10 avg pm2.5 avg	LYS1 lux cat LYS1 b' LYS1 g' LYS1 kelvin LYS1 lux LYS1 medi LYS1 movement LYS1 r' LYS1 rgbb LYS1 rgbg LYS1 rgbr LYS1 rgbr	LYS2 lux cat LYS2 b' LYS2 g' LYS2 kelvin LYS2 lux LYS2 medi LYS2 movement LYS2 r' LYS2 rgbb LYS2 rgbg LYS2 rgbr LYS2 rgbr	kelvin avg lux avg medi avg	GVI 50m GVI grass 50m GVI phase mean GVI plant 50m GVI tree 50m Grass phase mean Plant phase mean Tree phase mean	Building % Combined Vegetation % Grass % Plant % SVF Sky % Tree % Wall %	band proportion 1000-4000 Hz band proportion 125-1000 Hz band proportion 20-125 Hz band proportion 4000-8000 Hz band proportion 8000-20000 Hz category dominant frequency (Hz) mechanisms prop mechanisms score octave proportion 1000 Hz octave proportion 125 Hz octave proportion 2000 Hz octave proportion 250 Hz octave proportion 31.5 Hz	octave proportion 4000 Hz octave proportion 500 Hz octave proportion 63 Hz octave proportion 8000 Hz rural natural prop rural natural score spectral centroid (Hz) spectral rolloff 85% (Hz) unidentify prop urban manmade negative prop urban manmade negative score urban manmade neutral prop urban manmade neutral score urban manmade positive prop	urban manmade positive score vehicle prop vehicle score wind prop wind score	Clearness Index Clearness Index G0 Clearness Index Infer	acc x acc y acc z eda phasic eda scl usiemens eda tonic heart rate hrv fd hf hrv fd hfnu hrv fd lf hrv fd lf hf ratio hrv fd lfnu hrv fd total power hrv fd vlf	hrv td rmssd hrv td sdnn met skin temperature vector magnitude	blink duration s distance from center eyelid aperture avg fixation duration s fixation rate gaze centrality gaze dispersion gaze velocity pupil change rate pupil diameter avg saccade amplitude saccade duration s saccade peak velocity stress composite	current comfort 1 7 environmental comfort mean 1 5 pleasantness 1 5 prs11 being away mean 1 5 prs11 coherence mean 1 5 prs11 fascination mean 1 5 prs11 scope mean 1 5 prs11 total mean 1 5 space familiarity 1 5 stai6 mean 1 5 thermal comfort 1 5	Date Date (Atmo/LYS) Date (UCM) Date (eye) Datetime GPS HDOP GPS alt GPS hdg GPS lat GPS lon GPS speed valid sample count GVI match dist m HRV 5min available HRV 5min max gap (s)	HRV 5min removed pct HRV 5min segment end HRV 5min segment start HRV 5min valid ml count HRV 5min valid span (s) MAG hdg ParticipantID PhaseID age bike frequency bmi category who bmi kg m2 clothing accessories clothing footwear	clothing height raw clothing lower body clothing skin tone clothing upper body clothing weight raw commute modes daily walking end time s gender has children height m name nationality own bike for experiment	participant id pct complete phase start offset s regular physical activity residence type residential area response index segment site familiarity 1 10 vision correction weight kg years in brussels
STEP 2 **Representatives (16)** **Rules + coverage**	AIR RH AIR temp HUMIDEX	MRT PET SND dBA SUN Gh	UTCI WIND AWS		pm1 avg pm10 avg pm2.5 avg			kelvin avg lux avg	GVI 50m	SVF												

STEP 3 Domain Grouping (16)	Thermal	Index	Acoustic	Radiation	Pollution	Light	Greenery
	AIR temp AIR RH WIND AWS	HUMIDEX PET UTCI	SND dBA	MRT SUN Gh SVF	pm1 avg pm2.5 avg pm10 avg	lux avg kelvin avg	GVI 50m

STEP 4 Equations + design VIF (12 exploratory candidates)	
Eq1: AIR_temp + AIR_RH + WIND_AWS + PM1 + (AIR_temp + AIR_RH + WIND_AWS) x sound + MRT x sound + common terms	max VIF = 2.53
Eq2: AIR_temp + AIR_RH + WIND_AWS + PM2.5 + (AIR_temp + AIR_RH + WIND_AWS) x sound + MRT x sound + common terms	max VIF = 2.53
Eq3: AIR_temp + AIR_RH + WIND_AWS + PM10 + (AIR_temp + AIR_RH + WIND_AWS) x sound + MRT x sound + common terms	max VIF = 2.54
Eq4: HUMIDEX + PM1 + (HUMIDEX) x sound + MRT x sound + common terms	max VIF = 2.48
Eq5: HUMIDEX + PM2.5 + (HUMIDEX) x sound + MRT x sound + common terms	max VIF = 2.48
Eq6: HUMIDEX + PM10 + (HUMIDEX) x sound + MRT x sound + common terms	max VIF = 2.48
Eq7: PET + PM1 + (PET) x sound + MRT x sound + common terms	max VIF = 3.02
Eq8: PET + PM2.5 + (PET) x sound + MRT x sound + common terms	max VIF = 3.02
Eq9: PET + PM10 + (PET) x sound + MRT x sound + common terms	max VIF = 3.02
Eq10: UTCI + PM1 + (UTCI) x sound + MRT x sound + common terms	max VIF = 3.05
Eq11: UTCI + PM2.5 + (UTCI) x sound + MRT x sound + common terms	max VIF = 3.05
Eq12: UTCI + PM10 + (UTCI) x sound + MRT x sound + common terms	max VIF = 3.05

Common main terms: sound + MRT + SUN_Gh + SVF + lux + kelvin + GVI + exertion + Phase (sum-to-zero). All interactions retain their main effects.

Rank: 312/312 designs full rank; 1328-2057 complete rows. Exposures: bout-centered, scaled by pooled SD; participant is the independent cluster.

VIF: maximum across 26 outcome-specific designs, including interactions and grouped PhaseID; >=5 is a review flag, not automatic deletion. Candidate equations are NOT fitted results.

STEP 5: fit the specified equations, assess covariance/residuals and multiple testing. Previous coefficients belong to the previous four-equation specification.